Computers and Education: Artificial Intelligence 10 (2026) 100588

Contents lists available at ScienceDirect

Computers and Education: Artificial Intelligence

journal homepage: www.sciencedirect.com/journal/computers-and-education-artificial-intelligence

Generative AI in high school English career preparation units: Student interactions, perceptions, and ethical concerns

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ARTICLE INFO

Keywords:
Generative artificial intelligence
High school
AI-assisted learning
Student–AI interaction
Career preparation
Learner agency

ABSTRACT

The rapid integration of generative AI into K–12 education presents both opportunities and challenges for supporting students' learning. This mixed-methods case study examined how high school students interacted with an AI-enhanced resume writing unit and a text-based mock interview unit within an English Language Arts course. Quantitative results indicated students' positive perceptions of AI utility across constructs of awareness, usage, evaluation, and ethics. Qualitative analysis revealed that students used AI to explore career information, generate and refine resume content, and iteratively revise and refine interview responses. While many students engaged critically—questioning, personalizing, and revising AI suggestions—others adopted a more passive or overly dependent approach. Students reported feeling more confident and valued individualized feedback, yet they also identified challenges such as AI's generic or decontextualized responses, difficulties interpreting AI feedback, and ethical concerns regarding authenticity, plagiarism, and overreliance. This study argues that meaningful integration of generative AI in secondary classrooms requires explicit instruction in prompting, verification, and critical evaluation so that students develop both the agency and the skills necessary to use AI as a scaffold rather than a substitute for learning.

1. Introduction

The integration of artificial intelligence (AI) into high school education presents both promising opportunities and significant challenges. Recent studies highlight several potential benefits of AI in educational settings: it can monitor students' learning progress, deliver personalized guidance based on real-time assessments, respond promptly and accurately to student inquiries, and even act as a co-constructor of knowledge (Law, 2024; McKnight, 2021; Sutjarittham et al., 2019). These affordances suggest that AI can enhance efficiency, engagement, and differentiation in learning.

However, these benefits are accompanied by valid concerns. Overreliance on AI tools may lead to superficial learning or passive engagement, where students bypass the cognitive processes essential for deep understanding (Li et al., 2024; Zawacki-Richter et al., 2019). Additionally, students may inadvertently absorb biases embedded in AI-generated content, face risks to privacy and data confidentiality, or engage in unethical academic practices (Chan & Hu, 2023). Thus, the educational community is faced with a pressing question: *how can we maximize the educational value of AI while minimizing its potential harms?*

Rather than advocating for or against AI integration, this study seeks to identify effective educational strategies that amplify the benefits of AI and mitigate its limitations in high school settings. Specifically, we examined students' authentic learning experiences in a unit on career preparation, focusing on how they interact with AI tools and what insights can be drawn to inform teaching and learning.

A central consideration in this inquiry is how students utilize AI during the learning process. The literature emphasizes the value of active learning, which involves engaging students in thinking, discussing, investigating, and applying knowledge (Michael, 2006). From this perspective, the nature of students' interaction with AI, whether passive or participatory, can greatly influence learning outcomes. For example, asking AI to generate an entire essay may inhibit learning, while using AI to receive constructive feedback can foster critical reflection and revision, ultimately leading to deeper understanding. Therefore, the intentionality behind AI use becomes a crucial determinant of its overall educational impact.

Closely related is the concept of AI literacy, particularly students' ability to formulate effective prompts. Prompt engineering, which is referred to as a skill of crafting queries that yield meaningful outcomes

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https://doi.org/10.1016/j.caeai.2026.100588
Received 1 December 2025; Received in revised form 26 March 2026; Accepted 27 March 2026
Available online 28 March 2026

from AI, plays a vital role in the quality of AI interaction (White et al., 2023). Just as the effectiveness of surgery depends more on the surgeon rather than the scalpel, the educational benefit of AI depends on the user's skill. If students lack the competencies needed to engage AI meaningfully, its use may hinder rather than enhance learning. Despite the importance of these skills, research on what constitutes AI competency and how it can be taught remains limited.

Further, students' perceptions of AI play a critical role in adoption and effective use. Drawing on models such as the Technology Acceptance Model, factors like perceived usefulness and ease of use are strong predictors of students' willingness to adopt emerging technologies (Ma, Pengcheng, et al., 2025). In the context of AI-supported learning, these perceptions influence whether students view AI as a meaningful learning support tool and whether they are willing to engage with it during learning activities. If students view AI as irrelevant or burdensome, they may resist using it even when it has the potential to support learning. Therefore, it is essential to understand how students experience and evaluate AI through both cognitive and affective lenses. Indeed, these perceptions are multifaceted, shaped by individual, social, and contextual factors, such as prior experience, peer influence, and the ways in which AI is integrated into the curriculum (Acosta-Enriquez et al., 2025; Gerlich, 2023).

Building on these considerations, the current study aims to examine high school students' interactions with and perceptions of AI in a real-world classroom context. Specifically, we focus on students' participation in career development units within an English class, where they used AI tools to develop resumes and practice text-based job interviews. Research on AI in writing tasks has primarily focused on writing performance (e.g., Liu et al., 2023; Zhao, 2025) and learners' perceptions (Wang, 2025), reporting both positive impacts and limitations of AI use. The findings from such work highlight the importance of (a) effective learner-AI interaction and (b) fostering positive perceptions of AI for its meaningful use. As such, we seek to address two key research questions:

1. How do students interact with AI while learning career development units?
2. How do students perceive the utility of AI for their learning?

By analyzing students' lived experiences with AI-supported learning activities, this study contributes to the growing body of research on AI in education. Specifically, it aims to inform the development of pedagogical strategies, curriculum design, and teacher professional development that promote both innovation and academic integrity in secondary education.

2. Literature review

2.1. AI-supported writing and feedback

Given the potential benefits and challenges of using AI in education, exploring its application specific to subject matters is crucial. This is because the successful use of a technical innovation for teaching and learning depends on the understanding of not only how it works but also how it can facilitate student learning and teachers' pedagogical approach in a disciplinary classroom context (Mishra & Koehler, 2006). One of the key areas worth exploring in this regard is writing, which is considered a fundamental skill to support cognition (Kellogg, 2008). Writing can engage learners in analyzing and organizing information and synthesizing ideas into a coherent structure. Thus, writing is not just about a reflection of thoughts but also an active process of refining initial thinking via planning, drafting, receiving feedback, and revising (Flower & Hayes, 1981). Because this process can create a significant cognitive load, learners often require scaffolding to navigate its essential stages effectively (Evmenova et al., 2024).

To support the evaluation of students' writing tasks and to deliver timely, individualized feedback, researchers have increasingly explored the use of Automatic Writing Evaluation (AWE) systems such as Grammarly (Huawei & Aryadoust, 2023; Steiss et al., 2024). These systems provide feedback tailored to each learner's progress and enable multiple rounds of revision, which has been shown to enhance both writing quality and student engagement (Zhai & Ma, 2022). Nevertheless, AWE tools also present notable limitations: their feedback can sometimes be confusing or less accurate than that of human raters, and their initial setup may require substantial time and effort (Shermis, 2014).

The emergence of large language models (LLMs), a form of generative artificial intelligence (GenAI) exemplified by tools such as ChatGPT, has ushered in a new era for AWE systems (Escalante et al., 2023). Unlike earlier systems, these models can not only detect and suggest corrections for common grammatical and stylistic issues (Marzuki et al., 2023) but also generate coherent sentences, paragraphs, and even complete essays (Law, 2024). Most notably, GenAI tools enable dialogue-based interactions by producing human-like responses to user queries (Evmenova et al., 2024). Such interactive exchanges and generative capabilities offer valuable opportunities for providing timely, personalized feedback for assisting writers throughout key stages of the writing process, including planning, drafting, reviewing, and revising (Graham et al., 2014; Steiss et al., 2024).

Research examining feedback from GenAI tools in writing tasks has primarily focused on outcomes related to writing performance and motivation (e.g., attitudes). Findings regarding writing performance, however, have been mixed. On the positive side, Chan et al. (2024) investigated the effects of LLM-based feedback on first-year college students' essays and found that students who received revision feedback from the LLM system demonstrated significantly greater improvements in their writing than those who did not. Similarly, Lin et al. (2025), in a study involving high school students' compositions, reported that a GenAI system providing feedback on coherence, style, and grammar not only enhanced writing quality but also fostered deeper reflection during the writing process.

Despite these positive outcomes, concerns have also been raised regarding the impact of GenAI tools in writing contexts. Several potential issues associated with ChatGPT, for example, include superficial engagement with content, overreliance on AI-generated outputs, and reduced opportunities for developing critical thinking skills (Bai et al., 2023). Supporting these concerns, a recent study examining brain activity during adult participants' interactions with ChatGPT in writing tasks found evidence of limited neural connectivity, suggesting reduced cognitive engagement and internal idea generation (Kosmyna et al., 2025).

The nature of learners' interaction with GenAI tools—particularly the type of feedback sought and generated—appears to influence these outcomes. For instance, in a recent study by Lee and Son (2024), high school students received GenAI-generated feedback on their drafts. While participants showed improvements in grammar, no significant gains were observed in the content quality of their writing. The authors attributed this to the nature of the prompts that students used, which predominantly requested suggestions on vocabulary, grammar, and mechanics rather than on content development. Similarly, in a study by Yin et al. (2025), college students received feedback from an AI-based chatbot on their understanding of the human cardiovascular system. The findings revealed that affective feedback (e.g., encouraging statements) and neutral feedback (e.g., non-evaluative or non-judgmental statements) were less effective than metacognitive feedback (e.g., prompts encouraging self-regulation) in promoting conceptual understanding.

2.2. Attitudes toward AI-generated feedback in writing

Similar to findings on GenAI's impact on writing performance and learning, research has reported mixed results concerning students' attitudes toward AI-generated feedback. In a study by Chan et al. (2024), LLM-based feedback was found to generally enhance students' motivation for essay writing; however, some participants reported feeling disengaged or uninspired. The authors suggested that the multiple iterations of feedback generated by the LLM system might have been cognitively or emotionally taxing for students. Other studies have compared AI-based feedback with that provided by human experts. For instance, Escalante et al. (2023) examined feedback on college students' weekly essays delivered by both a human tutor and ChatGPT. Approximately half of the students preferred AI-generated feedback, perceiving it as more specific and clearer, whereas the remaining students favored human feedback due to the more engaging and personally meaningful nature of face-to-face interaction.

Comparative studies have also explored how expert raters and educators evaluate the quality and usefulness of AI-generated feedback in relation to human feedback. For instance, Behzad et al. (2024) analyzed student essays from a community writing forum in which AI-generated feedback was compared with feedback provided by human community members. Expert evaluators rated the human feedback as superior in most categories—including relevance, identification of weaknesses, specificity, and helpfulness—while the AI-generated feedback was judged more effective only in addressing strengths. In a separate study, Jansen et al. (2024) asked preservice teachers to evaluate secondary students' essays accompanied by two feedback messages: one generated by an LLM and the other written by an expert teacher. The majority of participants rated the expert teacher's feedback as more useful, attributing this preference to its clearer and more straightforward language.

2.3. Learner-AI interaction and perception of AI feedback

The implications of the aforementioned studies on AI use in writing tasks highlight two key considerations: (a) the importance of effective learner–AI interactions through well-crafted prompts and meaningful feedback, and (b) the necessity of fostering positive perceptions of AI to support its meaningful adoption. As demonstrated in Lee and Son's (2024) study, the prompts that students use when interacting with GenAI tools can influence the focus of the generated feedback, which in turn may affect the quality and organization of their revised content. Moreover, the nature of the feedback—shaped in part by the prompts—appears to influence students' levels of cognitive engagement (Kosmyna et al., 2025). Consequently, further research is needed to develop strategies that help students craft effective prompts for generating feedback that meaningfully supports the writing process.

In addition, as several studies have shown, despite some positive motivational effects, feedback from GenAI tools is often perceived as less useful than feedback provided by human experts (e.g., Behzad et al., 2024). This perception is important because individuals' beliefs about both their competence in a given task and the relevance or usefulness of that task influence their future engagement (Wigfield & Eccles, 2002). Indeed, students' perceptions of AI's usefulness and ease of use have been found to predict their willingness to engage with such tools (Ma, Wang, et al., 2025). Therefore, understanding how students experience and perceive GenAI tools is essential for identifying and developing strategies that effectively support their learning with these emerging technologies.

3. Method

3.1. Research design

This research employed a single-case design with multiple embedded units of analysis (Yin, 2018). The overarching case examined was students' experiences in an AI-integrated high school English Language Arts (ELA) class, with two embedded units of analysis: (a) resume writing and (b) mock interview. A mixed-methods concurrent design guided the study, integrating qualitative and quantitative data collected simultaneously to provide complementary perspectives on students' experiences (Leavy, 2022). Consistent with the case study approach, the qualitative data served as the primary source of insight, while the quantitative survey results were used descriptively to contextualize students' perceptions of AI within this instructional setting.

3.2. Participants and learning contexts

Fifty-five 10th graders, across three sections of an ELA class taught by an English teacher, participated in the study. The high school was located in the U.S. Midwest and classified as a town locale, a category describing schools situated in small urban clusters outside major metropolitan areas. Although classified as a town, the school is situated within a broader rural Midwestern region, where socioeconomic conditions and digital infrastructure may differ from those in urban or suburban areas. All participants submitted consent forms prior to the start of the study, and both the researchers and the teacher adhered to Institutional Review Board (IRB) protocols approved by the researchers' institution.

The researchers selected the career preparation units as the focal instructional sequence because it requires students to develop resumes and practice interview skills, which involves iterative drafting, revision, and practice. These professional genres also lend themselves well to AI-supported learning, as AI systems can provide immediate feedback, generate alternative phrasing, and simulate interactive interview scenarios. Considering these features, the units were designed to provide authentic learning contexts in which students could interact with AI tools during career preparation activities.

The teacher and the researchers collaboratively designed lesson plans that integrated a generative AI system based on LLM to support these skills. Specifically, the study utilized the School AI platform (powered by GPT-3.5), which provides enhanced data privacy compared to standard LLM interfaces. AI models were tuned to align with the learning objectives and classroom activities so that student–AI interactions were pedagogically relevant. In particular, the AI system was trained with step-by-step prompts that define its role, objectives, and activities, incorporating example questions that students would ask as well as a clear feedback format (see Appendix A). The system was further designed to prompt follow-up questions and encourage students to elaborate on their responses, enabling more personalized feedback and iterative refinement of their drafts. The AI tools were accessed through the School AI platform, for which the school district held a license. Although the units extended beyond the AI-integrated lessons, researchers collected learning data only from the classes in which students interacted with the AI. Each class session lasted 45–50 min, and students worked individually on Chromebooks.

The AI-enhanced resume unit spanned two days. On Day 1, students drafted their resumes by reviewing a sample resume from a former student and using AI to brainstorm ideas and gather information. On Day 2, students revised and refined their resumes with AI support.

The AI-enhanced mock interview unit took place over one day. Prior to this lesson, students had learned about basic job interview procedures and common interview questions. During the AI-integrated lesson, students participated in a text-based mock interview in which the AI posed questions and students typed their responses. The AI then provided feedback on each response, including strengths, areas for improvement, and example answers.

The study did not include an explicit prompting framework. However, the instructional materials functioned as embedded prompting guidance. In the resume unit, students drafted their own resumes prior to AI interaction. These drafts served as the primary input for AI-supported revision, thereby structuring context for their interaction. Similarly, in the mock interview unit, students were provided with interview questions in advance of their AI-based mock interviews. Although these drafts were not explicitly introduced as prompting strategies, they operated as indirect scaffolds that shaped how students formulated their responses when interacting with AI and potentially influenced how they approached follow-up queries. After each lesson, students completed an electronic worksheet reflecting on their experience using AI (See Appendix B). Detailed lesson plans for both units are provided in Appendix C.

3.3. Data collection

Primary data sources included: (1) student–AI interaction logs; (2) students' written reflections for each lesson; and (3) 5-point Likert-scale post-surveys. Interaction logs were collected from 48 students for resume drafting, 47 for resume refinement, and 49 for the mock interview. Written reflections were obtained from 29 to 47 students for the resume unit and the mock interview unit, respectively.

Each reflection set included six questions in a consistent format, addressing topics such as students' interpretations of AI feedback and refinement processes, challenges encountered, confidence, key takeaways, and ethical considerations. Example items included: "What was the most helpful AI suggestion for improving your resume? Why?", "What challenges did you face when using AI for resume refinement?", and "What ethical concerns, if any, do you have when using AI for resume writing or schoolwork?"

Post-survey data were collected from 39 students for the resume unit and 49 students for the mock interview unit. The survey items were adapted from B. Wang et al. (2023)'s work, which assessed AI awareness, usage, evaluation, and ethical considerations. The core constructs and item structure were retained, with minor contextual modifications to reflect the resume and mock interview activities (see Appendix D).

Supplementary data sources included: (1) students' reflections on the overall career preparation units, (2) researchers' field notes, and (3) brief post-lesson interviews with the teacher—approximately 20 min for the resume lesson and 10 min for the mock interview lesson.

3.4. Data analysis

For quantitative analysis, descriptive statistics and reliability analysis on survey items were conducted using R 4.40. For qualitative analysis, thematic analysis was employed to identify, analyze, and report patterns (Braun & Clarke, 2006). Codes and themes were created inductively from the raw data. One researcher created initial codes and themes in MAXQDA 24. These findings were subsequently reviewed and refined through three rounds of collaborative discussions among all researchers. To facilitate this process, shared Microsoft Excel and Word documents were utilized to document discussions.

4. Results

4.1. How do students interact with AI in the career development units? (RQ1)

4.1.1. Students' interactions with AI in the resume unit

The analysis identified four themes that captured students' interaction with AI in the resume unit: (1) exploring career information, (2) generating resume content, (3) seeking feedback and revising drafts, and (4) refining through iterative dialogue with AI. These themes highlight how students utilized AI while learning to write professional resumes (see Table 1).

4.1.1.1. Exploring career information. At the initial stage of the resume unit, students often used AI to gather basic information relevant to their career interests. This included exploring relevant certifications, professional memberships, and both paid and volunteer experiences that could strengthen their resumes. Many students interacted with AI to gain insights into the key traits, skills, and qualifications required for jobs. These interactions not only supported their information gathering but also helped them better understand the expectations of the job market and what employers look for in specific roles.

Table 1
Emergent themes of Student-AI interactions in the resume unit.

Themes	Description	Examples
Exploring career information	Students used AI to explore career options, job descriptions, and professional expectations.	"Explain someone who would fit the sales manager role perfectly." "What are so hard skills required for travel agents?" "Can you go to OO University for ultrasound tech?"
Generating resume content	Students prompted AI to draft specific sections or full versions of their resumes.	"Help me write my professional summary." "Give me a professional summary of someone that fits the role of a sales manager."
Seeking feedback and revising drafts	Students sought AI feedback to identify issues and guide revisions.	"I am looking at being a travel agent. Can you suggest improvements on my resume?" "Is there any other ways I can improve my Professional Summary for a Travel Agent?"
Refining through iterative dialogue with AI	Students engaged in iterative conversations to improve resumes, involving both surface-level and deep interactions.	Students who interacted with AI back and forth by asking and answering questions, encompassing a spectrum of engagement from surface-level to deep interactions.

Some students were able to clearly express their intentions and engage in meaningful, goal-oriented interactions with the AI. For instance, one student started an interaction with AI, saying, "I'm interested in applying for a corporate pilot." Then, the AI asked any relevant experiences and qualifications that he had, followed by him sharing his familiarity with pilot work, as his dad works as a pilot, "My dads a pilot so he can help me get experience", and further sharing his learning experience with pilot job, "I have shadowed him at work", "I learned how to do invoices." Such interactions demonstrated that students gathered basic information relevant to their goals by sharing their relevant experiences.

The students also became aware of the limitations of AI in providing up-to-date information, which led them to question the reliability of AI-generated responses. For example, when one student asked, "Can you go to OO University for ultrasound tech?", the AI replied, "I cannot provide real-time information about specific programs like the Ultrasound Technician program at OO University. However, I can help you make your resume stand out for this field!" This interaction reminded students that AI can support information seeking, but only when used with critical thinking. Students can search for and synthesize information with AI assistance, but they must also question the accuracy of AI outputs and recognize the need for additional verification from other sources.

4.1.1.2. Generating resume content. Most students utilized AI to draft brief bullet points describing their previous experiences, write comprehensive professional summaries, and develop subsections such as skills and education in their resumes. As the unit progressed, students increasingly relied on AI to generate full sections of their resumes from scratch, using AI-produced drafts as a foundation. They then refined these drafts by incorporating their own details, drawing on AI suggestions as anchor points to make the content more accurate and personalized.

For example, one student who prepared a resume for an athletic trainer started drafting his professional summary section by saying, "Help me write my professional summary." In response, the AI outlined the key components that such a summary should include, such as relevant skills, experience, and career goals, rather than providing complete answers. Some students, however, initially chose to generate complete resume content using AI. For instance, one student asked, "Give me a professional summary for someone who fits the role of a sales manager," and the AI generated a full professional summary.

4.1.1.3. Seeking feedback and revising drafts. Beyond initial drafting, students also used AI to revise and refine their resumes according to AI feedback. Rather than passively accepting AI generated content as is, many students engaged deliberately with the AI, asking focused questions to invite critiques and revise their drafts. This indicates a shift from basic content generation to more intentional, reflective, and critical engagement with the writing process.

Students sought feedback typically in one of three ways: asking general questions, requesting feedback on specific sections, or seeking alternative word choices or phrasing. Some students requested broad suggestions to improve overall resume quality. For example, one student asked, "I am looking at being a travel agent. Can you suggest improvements on my resume?" This illustrates students' openness to general enhancement and their recognition of the resume as an evolving document.

Students also asked AI for feedback on specific sections. For instance, some focused on particular components, such as asking, "Is there any other ways I can improve my Professional Summary for a Travel Agent?" This example reflects students' targeted engagement and their willingness to revise based on detailed input. Further, students sought improved phrasing or word choices. Requests like "Can you word the example for 'Mention Relevant Experience' differently?" demonstrate their attention to tone, clarity, and professionalism, suggesting an effort to achieve both content precision and rhetorical effectiveness.

However, not all uses of AI demonstrated such intentional engagement. In some cases, students relied heavily on the tool without providing clear directions or contextual information. For instance, some students simply copied and pasted their content with minimal explanation, which often resulted in generic or vague feedback from the AI. One student even prompted, "Can you give me an edited version?" without offering any personal input—illustrating a more passive interaction. Although such overreliance may still produce useable output, it can limit opportunities for self-reflection, critical thinking, and the development of independent writing skills.

4.1.1.4. Refining through iterative dialogue with AI. Students engaged in conversations with the AI to refine their resumes in a back-and-forth manner, treating the tool as a collaborative partner. They used AI primarily to elaborate on ideas by receiving feedback and refining their content iteratively. This process often began with students submitting an initial draft or idea and then evolved through AI-generated prompts and students' follow-up questions. This process encompasses a spectrum of engagement from surface-level to deep interactions.

Primarily, surface-level interactions were characterized by passive engagement, such as providing brief responses to AI prompts and uncritically accepting AI-generated suggestions. For instance, one student responded to an AI query regarding professional qualities by stating, "I fixes cars at my house, and I change oil and change breaks and all that," followed by, "and I change tires." Upon receiving the AI-composed summary, the student immediately terminated the interaction. Similarly, another student initiated a dialogue by stating, "I would like to start working on my professional summary," and answered follow-up questions about strengths with, "some strengths that define me are hardworking and a problem solver." Once the AI generated the professional summary, the student transitioned to the next section of the resume without further refinement.

Conversely, deep interactions extended beyond passive acceptance, characterized by active inquiry and iterative refinement prompted by AI suggestions. For instance, one student began the conversation by describing his experiences in elementary school sports clubs and the lessons he learned from them. In response, the AI suggested a refined version, organizing it with subsections and bullet points, and asked a follow-up question: "Do you have any specific achievements or responsibilities from your volunteer work or sports that you'd like to highlight? This could make your section even stronger!" The student then revised his volunteer and club experience section and proceeded to his work experience, describing his tasks as a roofer and asking for suggestions on how to improve it further. The AI responded by recommending the use of stronger action verbs and clearer descriptions of his contributions. This example illustrates how iterative dialogue can facilitate the development of meaningful content, with each section being articulated and refined as the conversation progresses.

Notably, in some cases, students did not merely follow the suggestions from AI but demonstrated agency by questioning, exploring alternative ideas, and redirecting AI-generated responses to better align with their intentions. When the AI asked about a student's work experience to initiate resume writing, one student replied, "What if I've never worked a day in my life?" prompting the AI to offer alternative strategies for developing the resume. Additionally, when the AI sought feedback after refining a section by asking, "How does it sound?", one student requested a revision, saying, "Instead of the title and bullet point, can we make it just student teacher volunteer?" These conversations indicate that students demonstrated agency throughout the process, actively refining ideas and tailoring their resumes to align with their individual goals.

4.1.2. Students' interactions with AI in the mock interview unit

The analysis identified two themes that captured students' interactions with AI during the mock interview unit: (1) initial responses to AI interview questions and (2) reactions to AI feedback (see Table 2).

4.1.2.1. Initial responses to AI interview questions. During the AI-integrated interview practices, students' initial responses generally fell into four categories: (1) general and vague responses, (2) addressing questions without providing specifics, (3) limited connection to job qualifications, and (4) connecting personal experiences to job qualifications.

First, most students answered the questions in general or vague terms, and AI provided scaffolding to help them offer more specific and concrete examples. For instance, in response to the question, "Tell me about yourself," many students gave simple answers such as, "I am 16 years old. I play two sports, which have taught me hard work and dedication." AI typically acknowledged that their answers were engaging but encouraged them to elaborate with a concrete story or example related to their statements.

Second, some students addressed the questions and introduced relevant ideas, but their responses remained surface-level rather than detailed. For example, one student connected their caring nature as a personal strength to the role of a pediatric sonographer, stating, "My greatest strength would be my passionate care of children … Being caring toward children is important because you want the child to feel safe … they will be willing to come back to get the help they need." However, the response did not include a personal or situational example to illustrate this trait.

Third, many students' responses were not sufficiently aligned with the job qualifications, particularly when answering more challenging interview questions such as those about weaknesses, handling stress, or responding to criticism. While they addressed the questions, their answers often lacked connections to professional growth or motivation for the role. For instance, one student described their strategy for managing stress, saying, "I handle stress by breathing in and out in a slow way, usually I do 6 breathing in, 6 holding it, and 6 breathing out … if I'm helping someone all my stress goes away." Although the strategy was explained clearly, the response did not connect this coping method to the responsibilities or expectations of a social worker.

Lastly, some students shared personal experiences that effectively demonstrated their qualifications for the job. For example, one student

Table 2
Emergent themes of Student-AI interactions in the mock interview unit.

Themes	Subthemes	Descriptions	Examples
Initial responses to AI interview questions	General and vague responses	Students answered the questions in general or vague terms.	"I am careful and attentive. I finish things once I start them."
	Addressing questions without specifics	Students addressed the questions and shared relevant ideas, but their responses often remained general rather than specific.	"My greatest strength would be my passionate care of children … Being caring toward children is important because you want the child to feel safe … they will be willing to come back to get the help they need."
	Limited connection to job qualifications	Students addressed the questions but failed to connect their responses to professional qualifications or growth.	"I handle stress by breathing in and out in a slow way, usually I do 6 breathing in, 6 holding it, and 6 breathing out … if I'm helping someone all my stress goes away."
	Connecting personal experiences to job qualifications	Students presented personal experiences relevant to job qualifications.	"I am a very skilled individual with extensive knowledge in programming languages such as Python and Java."
Reactions to AI feedback	Ignoring AI feedback	Students moved on to the next questions without revision.	"Next."
	Accepting AI suggestions without reflection	Students accepted AI-suggested revised responses without critical consideration.	Students directly copied and pasted AI-suggested revised responses.
	Improving responses based on AI feedback	Students sought AI advice on interview strategies or improved their responses through iterative practice.	"How do I turn a negative question to a positive?" or Students brought specific examples to a question based on AI feedback.

applying for a software developer position shared their skills in programming languages and their passion for problem-solving and collaboration, saying, "… I am a very skilled individual with extensive knowledge in programming languages such as Python and Java … I am enthusiastic about applying problem-solving skills and collaborating with peers to create effective software applications …" This response clearly linked personal experience to the competencies expected in the role.

4.1.2.2. Reactions to AI feedback. Students' reactions to AI feedback fell into three categories: (1) ignoring AI feedback, (2) accepting AI suggestions without reflection, and (3) revising responses based on AI feedback.

First, many students ignored AI feedback and moved on to the next question without revising their answers. The AI was designed to provide feedback by highlighting strengths, areas for improvement, and suggesting revised responses. It then encouraged students to revise their answers before proceeding to the next question. However, many students skipped this optional revision step by simply responding, "Next," and continuing the interview without making changes.

Second, some students accepted AI-suggested revisions without engaging in critical evaluation. For example, one student responded, "I prefer to work as a team because I feel like I am easy to work with and a good listener," when asked, "Do you prefer to work independently or as a team?" The AI suggested adding a specific example from their sports or club experiences and even provided a model response. The student then copied and pasted the AI's revised response verbatim and proceeded to the next question, demonstrating limited reflection.

Third, a few students sought AI advice on interview strategies or meaningfully revised their responses through iterative practice. For example, one student asked AI how to answer negative questions—such as those about weaknesses or criticism—by saying, "How do I turn a negative question into a positive?" Another student requested targeted feedback on a specific answer, asking, "Can you give me some feedback on my answer to the last question?" These students also revised their responses based on AI suggestions. For instance, one student initially answered a strengths-related question by saying, "My greatest strength, like I said previously, is that I work well with others, and when I am trained properly, I always want to help." After AI recommended including a specific example, the student revised the answer to: "… One time, I helped a classmate in a science lab who did not know what they were doing. I showed them the correct way to align the atoms, and they learned, which helped us complete the project faster." This revision demonstrates the student's willingness to incorporate concrete evidence in response to feedback.

Additionally, there were nine mock-up interview questions in total, and a few students demonstrated more detailed responses over time, particularly in incorporating specific examples into their later answers. For instance, one student repeatedly received AI feedback such as "Consider sharing a specific instance …" for their early responses about strengths and weaknesses. By a later question—"How do you handle stress?"—the student applied this guidance, responding, "… When it comes to challenging situations, I just take deep breaths and try not to dwell on it. For example, recently my grandma passed away, and I had to remain calm for myself. I followed these exact steps." Although the example was more personal than professional, it demonstrated the student's effort to apply feedback by including a concrete example and using the phrase "for example" to structure their response.

4.2. How do students perceive the utility of AI for their learning? (RQ2)

Guided by the Technology Acceptance Model, the post-survey results provide descriptive insight into students' perceptions of AI's utility within this instructional context. Overall, students reported positive perceptions of AI across the constructs of awareness, usage, evaluation, and ethics (see Table 3).

Additionally, the analysis of students' open-ended written responses, along with field notes, and teacher interviews, identified five themes reflecting students' perceived utility of AI in the resume and mock interview units: (1) valuable AI suggestions for refinement, (2) increased confidence, (3) challenges, (4) ethical considerations, and (5) skepticism toward AI feedback.

4.2.1. Valuable AI suggestions for refinement

Students perceived AI's suggestions as valuable for refining their resumes and interview responses. First, they noted that AI helped

Table 3
Descriptive statistics of students' perceived utility of AI in resume and mock interview units.

Constructs	Resume ($n = 39$)		Mock Interview ($n = 49$)	
	M (SD)	Cronbach's α	*M (SD)*	Cronbach's α
Awareness	4.09 (.84)	.85	4.34 (.77)	.86
Usage	4.18 (.92)	.86	4.41 (.72)	.86
Evaluation	3.95 (.92)	.84	4.30 (.78)	.84
Ethics	4.05 (1.12)	N/A	4.37 (.83)	N/A

Note: Each construct was measured using three items, except for ethics, which consisted of a single item.

enhance the formal structure and professional tone of their writing. With AI feedback, students revised their work to use more formal and precise language, improve grammar, and strengthen clarity and conciseness. Many reported using more professional wording, revising sentence efficiency, and communicating their ideas more clearly.

Second, students perceived that AI helped them strengthen content relevance and personalization. They tailored their resumes and interview responses to the specific jobs they intended to pursue and practiced incorporating personal examples that directly addressed interview questions. The teacher also appreciated that AI could provide individualized feedback to many students simultaneously, remarking, "It's like having the individual teacher when I cannot get around to individualizing in that moment." These findings suggest that students found AI's guidance useful for enhancing the relevance of their responses through specific, personalized examples.

Lastly, students perceived that AI supported them in demonstrating self-awareness and reflective growth. With AI feedback, they engaged in critical reflection on their strengths and areas for improvement. One student, for example, noted the benefit of identifying areas for growth: "I liked that the AI gave feedback on what it liked while also showing what I could improve on." The teacher also emphasized the value of using AI to receive constructive criticism during class. These findings suggest that interactions with AI served as a meaningful opportunity for students to reflect on and refine their work.

4.2.2. Perceived increased confidence

Students perceived increased confidence after participating in the AI-integrated career preparation units. First, many reported feeling more confident in resume writing and mock interviews after completing the units. For example, students expressed confidence in the professional appearance of their resumes with AI support. They also viewed the mock interview session as valuable practice; as one student explained, "Yes, because it gave me a sort of dry run before the actual interview, so I knew what to expect." The teacher similarly emphasized how collaboration with AI fostered students' confidence, noting, "It furthers the discussion and builds confidence in a way that they can ask me more questions when they're more confident about their answers because it gave good constructive criticism and just complimenting too …"

Second, students recognized that they had developed strategies for effective resume writing and interview performance. They identified the importance of clear content, specific examples, professional language, and relevant connections to targeted jobs. One student, for instance, reflected on the value of aligning responses with a specific job field, stating, "Connecting all my responses to interior design will make me more likely to get the job." Students also viewed these activities as a checkpoint for identifying areas in need of further preparation for real job applications. These findings indicate that students gained important takeaways related to career-readiness strategies.

Lastly, students recognized AI as a valuable support tool for career preparation. Many expressed willingness to use AI in the future for resume writing and mock interview practice. They viewed AI as a resource for improving clarity and providing an additional perspective. As one student noted, "A good look from another perspective that could help improve my writing is helpful."

4.2.3. Challenges

Despite the aforementioned benefits, the students reported several challenges when interacting with AI. First, they noted a lack of contextual relevance in AI feedback. The feedback was often considered vague, general, or insufficiently personalized, and some responses felt overly robotic. For example, one student observed that "some of the suggestions were generic and lacked context or information," while another expressed discomfort with the robotic tone of AI-generated responses, stating, "AI made my responses sound too smart, almost inhuman-like." These findings indicate that students struggled when AI feedback did not align with their goals or felt disconnected from their personal context.

Second, students perceived difficulties related to the clarity and interpretability of AI feedback. They sometimes found it challenging to understand how to apply AI's suggestions or interpret what the system was asking for. To address this, students often requested more detailed explanations from the AI. One student highlighted the contrast between human–human and human–AI communication, noting, "There are certain ways you have to phrase things when speaking to AI compared to speaking to a person." These results suggest that although students found AI feedback difficult to interpret at times, they developed strategies to navigate and overcome these challenges.

4.2.4. Ethical considerations

Students perceived several ethical considerations surrounding AI use. First, they raised concerns about authenticity. Some worried that AI-generated content might misrepresent or exaggerate their abilities. For example, one student reflected on the possibility of overrepresenting oneself, stating, "If you use it too much, the company could think they are hiring a professional person, when really you aren't." Another student described discomfort with AI's unusual word choices and questioned the fairness of using AI in job applications, explaining, "My employer or teachers may notice that I use different words than I normally use … People who actually worked hard on it don't get the job because others used AI for their resumes and got the job over them." One student also doubted the authenticity of AI-generated revisions, noting, "It seemed like the AI wasn't truly reading my answer but was instead just picking out big words and suggesting I use fancier language for my resume." These findings indicate that students were aware of risks related to authenticity and misrepresentation.

Second, students expressed concern about becoming overly dependent on AI and, in turn, losing opportunities to think critically or learn independently. Several students noted a potential loss of learning, with one stating, "Students will just use it to get the answer and not to learn how to do it." Another student worried about relying too heavily on AI, remarking, "It changed our writing a little too much, and we used it too often." These results suggest that students recognized the potential reduction of personal agency and skill development associated with overreliance on AI.

Third, students identified concerns related to plagiarism and academic integrity. They expressed apprehension about copying AI-generated content directly, which could compromise originality. One student described direct copying as unethical, saying, "People could copy and make it be like their own, which isn't ethically right because it's not their own work." In the mock interview unit, students also raised concerns about cheating, with one stating, "Some people may find it cheating to use AI to help with preparation or academic work." These findings suggest that students were mindful of ethical issues surrounding plagiarism and academic integrity.

Lastly, some students reported having no ethical concerns regarding AI use. For example, one student stated, "I don't believe that there are any ethical concerns in using AI." Similarly, another viewed AI as a supportive tool, explaining, "If you use AI as a tool and not for your whole process, there is nothing wrong with it." These findings demonstrate that students varied in their perspectives on AI ethics, ranging from significant concern to full acceptance.

4.2.5. Skepticism toward AI feedback

While many students saw the utility of AI in their career preparation, a few expressed skepticism toward AI feedback. Some felt that AI offered only limited assistance or did not increase their confidence. For example, one student did not find AI useful for resume writing, stating, "Not really, because I feel that I had enough information already." Another student acknowledged that AI was somewhat helpful but did not boost their confidence, noting, "I really liked how it helped me improve my special skills, but it doesn't make me feel confident that it used AI pronouns in my summary." A student in the mock interview unit highlighted the gap between real-life and AI-assisted practice, explaining, “No, I have been through multiple interviews and they were nothing like that.” Another student negatively evaluated AI feedback, stating, “No, it didn't really help me that much. It gave me ideas and suggestions I didn't like.” These findings suggest that students varied in their acceptance of and perspectives toward AI feedback.

5. Discussion

The findings demonstrated that students engaged with AI in multiple, meaningful ways during the career preparation units. Consistent with the Technology Acceptance Model, students' generally positive perceptions of AI's usefulness appeared to support their willingness to engage with the tool during resume writing and interview preparation activities. Specifically, students used AI to explore career information, generate and refine resume content, and practice interview responses through iterative dialogue. However, the depth and quality of these interactions varied considerably, ranging from highly reflective and collaborative engagement to more passive or overly reliant use of AI-generated responses. This variation in interaction patterns suggests that students' motivation and sense of agency—concepts central to self-determination theory—play an important role in shaping how deeply students engage with AI-generated feedback. For example, some students demonstrated strong agency by questioning AI suggestions, personalizing responses, and refining their drafts iteratively, whereas others adopted AI revisions with limited context or modification, resulting in surface-level interactions. Despite these differences in interaction depth, students' overall perceptions of AI were largely positive, with many highlighting perceived experiences of greater clarity, confidence, and self-awareness during the learning process. At the same time, students reported several challenges, including generic feedback, difficulties interpreting AI suggestions, and ethical concerns related to authenticity and potential overreliance on the technology. Taken together, these findings illustrated both the pedagogical potential and the inherent limitations of AI as a supportive learning tool within this classroom context.

5.1. Beneficial aspects of utilizing AI

The findings suggest that when AI tools are tuned to instructional goals, they can effectively support students—especially those with limited prior experience in career preparation tasks such as resume writing and interviewing. Given students' low prior knowledge and the high cognitive load associated with unfamiliar professional genres, utilizing AI can provide structured scaffolding and tailored feedback that is responsive to individual learners' needs (Kalyuga, 2007). In particular, AI appeared to support divergent thinking by enabling students to brainstorm career options, explore relevant volunteer or work opportunities, and examine the skills required for specific professional pathways, which could enhance creative thinking in career preparation (Chang & Li, 2025). Students often experience cognitive fixation, relying on habitual thinking patterns that limit novel idea generation, when lacking domain-specific knowledge or experience (Chi et al., 1981). The AI tool acts as a collaborative partner that breaks this fixation by offering a vast range of potential responses drawn from extensive datasets. These diverse, AI-generated inputs serve as external stimuli that prompt students to look beyond conventional career trajectories. By engaging in an iterative dialogue—evaluating, refining, and building upon the AI's suggestions—students are pushed toward divergent thinking, ultimately synthesizing career pathways that are both novel and personally useful (Chang & Li, 2025). This form of guided exploration seemed to allow students to construct a clearer sense of direction in their career development in this study.

From the self-regulated learning perspective, this study also revealed how students can exercise strong agency during interactions with AI. While Vincoli et al. (2025) define agency as “a student's will and capacity to make decisions about their learning journey” (p. 220), its significance extends beyond simple decision-making; it determines whether AI limits learning to surface levels or facilitates deep learning. According to Yang et al. (2024), fostering this agency is crucial for enabling students to move from passive consumers of AI-generated content into active participants in knowledge creation. By engaging in reflective assessment—evaluating the utility and accuracy of AI outputs—students can extend their cognitive reach and integrate AI-driven insights into more sophisticated, self-directed reasoning (Dwivedi et al., 2023). Consequently, these findings challenge the concern that AI inevitably weakens cognitive development, which stems from overreliance on AI. Rather than passively accepting outputs, students in this study actively directed the dialogue by contextualizing their interests, refining their goals, and strategically narrowing the scope of information. This finding suggests that when students position AI as a collaborative partner in iterative exchanges, they maintain their agency and engage in practices that may foster deeper and more personalized learning (Kim & Lee, 2023; Roe & Perkins, 2024).

Further, this study highlighted that the students displayed varied interaction patterns, which might reflect their underlying intentions and motivations for using AI. While some students revised their resumes or interview responses thoughtfully in response to AI feedback, others accepted suggestions without reflection or ignored them altogether. The observed variability in student engagement with AI-generated feedback could be understood through the lens of self-determination theory, which posits that the quality of behavioral regulation depends on the innate psychological needs for competence, autonomy, and relatedness (Deci & Ryan, 2000). Specifically, students who thoughtfully revised their resumes demonstrated autonomous self-regulation, utilizing AI as a scaffold for mastery and deep cognitive processing. In contrast, those who accepted AI suggestions without reflection exhibited controlled regulation, prioritizing task efficiency over skill acquisition—a behavior characteristic of surface learning that risks cognitive offloading rather than genuine development (Mollick & Mollick, 2023). Consequently, because the specific type of regulatory process—whether autonomous or controlled—determines the quality of goal-directed behavior and the depth of conceptual processing, student motivation serves as the crucial predictor in learning contexts of whether AI integration leads to genuine skill augmentation or mere passive compliance (Deci & Ryan, 2000).

In addition to motivational factors, system-related perceptions may also influence students' engagement with AI-generated feedback. Some students in this study described the feedback as generic or lacking contextual relevance, which may relate to the concept of “information redundancy,” where repeated or predictable information reduces learners' cognitive engagement with AI (Wang & Sun, 2025). The effectiveness of AI-supported learning is fundamentally tied to the system's ability to generate context-sensitive and personalized guidance that aligns with a learner's specific intent, objectives, and abilities (Farhood et al., 2025). When feedback is repetitive or fails to adapt to unique student characteristics, it may lose its relevance and meaningfulness, potentially discouraging students from engaging deeply with the material.

In this study, the AI system was structured using prompts that defined its instructional role, objectives, and feedback format to align responses with the learning goals. While this helped ensure consistency in the feedback provided to students, it may also have resulted in responses that appeared repetitive or insufficiently personalized. Further, LLMs may produce more generalized feedback when students provide limited contextual information in their prompts or responses. Therefore, the perceived generic nature of the feedback likely reflects an interaction between the prompt design and the model's ability to generate context-specific guidance. This suggests that future classroom implementations may benefit from prompt designs that encourage students to provide richer contextual details while also leveraging advances in AI systems that support more personalized feedback.

5.2. Challenges to be considered

While the benefits of AI as a learning tool are evident, students' reflections revealed significant ethical and pedagogical tensions—particularly around the authenticity of their work and the risk of cognitive offloading. As Bastani et al. (2025) argued, while GenAI tools reduce task difficulty, it may paradoxically impede the development of the skills required to perform those tasks independently. Traditionally, students are positioned as primary agents in their learning, with teachers and peers serving facilitative roles. The introduction of GenAI, however, disrupts this dynamic. Many students bypass essential learning processes by soliciting direct solutions from AI, creating an illusion of knowing that ultimately undermines performance on authentic tasks (Bastani et al., 2025). This phenomenon represents a shift in AI's role from a scaffold to a replacement, producing outputs while circumventing the *productive struggle* essential for deep learning (Bauer et al., 2025).

This displacement of agency also raises profound concerns regarding authenticity. Students expressed discomfort that AI-generated artifacts might overrepresent their true abilities, creating a dissonance between their demonstrated performance and their internalized competence. When the gap between actual skill and AI-enhanced output becomes too wide, students may struggle to experience *psychological ownership* of their work (Pierce et al., 2001). Such diminished ownership not only invokes ethical questions surrounding academic integrity and plagiarism (Perkins, 2023) but also threatens students' self-efficacy (Rodríguez-Ruiz et al., 2025; S. Wang, Sun, & Chen, 2023). Because self-efficacy develops through mastery experiences, if students attribute success to the AI rather than their own effort, their confidence in the specific tasks may stagnate or decline, even when the final output is of high quality (Bandura, 1982).

The students also perceived that excessive reliance on AI could weaken their critical thinking skills, echoing scholarly concerns about cognitive offloading (Gerlich, 2025). A central question emerging from this study is therefore how to position students as critical evaluators and co-creators rather than passive recipients of AI-generated content. As previously discussed, motivation—whether autonomous or controlled—shapes how learners approach AI, yet motivation alone is insufficient without accompanying capability. Even highly motivated students may struggle to engage productively with AI if they lack the necessary "AI literacy," particularly the technical and critical competencies required to navigate generative technologies (Long & Magerko, 2020). The findings suggest that effective AI integration requires a dual-focus approach: fostering the will (motivation) to engage deeply and promoting the skill (AI literacy) to direct and interrogate the technology. Without explicit instruction on how to prompt, verify, and refine AI outputs, even well-intentioned students may fall prey to hallucinations or superficial engagement (Anders & Dux Speltz, 2025; Fan et al., 2025). Therefore, meaningful learning likely occurs only when students possess both the agency to direct the tool and the literacy to evaluate its contributions critically.

Another notable finding concerns students' skepticism toward AI feedback and their occasional decision to disregard AI-generated suggestions. While many students reported that AI feedback was helpful for refining their responses, some expressed doubts about its relevance or usefulness, which in certain cases did not contribute to increased confidence in the revision process. This skepticism may reflect students' attempts to evaluate the credibility and applicability of AI-generated responses within their own learning contexts. As the findings indicate, students sometimes struggled to recognize the relevance of AI feedback when a noticeable gap existed between their real-life experiences and the suggestions provided by the AI system. When the feedback appeared disconnected from students' personal contexts or learning goals, they were less likely to perceive it as meaningful guidance (Brandmo & Gamlem, 2025).

From a learning perspective, questioning and validating the credibility of AI-generated responses can represent a constructive practice for critical learners (Walter, 2024). Evaluating whether AI suggestions are accurate, relevant, or appropriate reflects an emerging form of critical AI literacy, in which learners actively interpret and assess AI-generated information rather than accepting it uncritically. However, the findings of this study also suggest that skepticism toward AI feedback may sometimes reflect negative attitudinal responses, particularly when students perceive the system as unhelpful, generic, or insufficiently aligned with their needs. Such perceptions may reduce students' willingness to engage with the feedback, leading them to dismiss or overlook suggestions rather than using them as opportunities for reflection and revision (Dey et al., 2025).

In this sense, students' skepticism toward AI feedback highlights a tension between critical evaluation and disengagement. While questioning AI-generated responses can support thoughtful learning and reflective judgment, excessive skepticism or a lack of perceived usefulness may limit students' motivation to interact productively with AI-supported feedback. These findings suggest that the effectiveness of AI-assisted learning depends not only on students' willingness to critically evaluate AI outputs but also on the system's ability to provide feedback that is sufficiently relevant, contextualized, and aligned with students' experiences and learning needs (Xu et al., 2025).

5.3. Implications for instruction and assessment

Given the concerns regarding disrupted student agency and over-reliance on AI, instructional interventions should be designed to position AI as a pedagogically structured co-creator rather than simply restricting its use in AI-integrated learning contexts. Teachers can emphasize transparency in AI use by requiring students to document when and where AI is used during the learning process, collecting records of students' interactions with AI such as prompt logs, and asking students to explain how they interacted with AI during the tasks (Rodger et al., 2025).

Teachers can design scaffolded AI interactions that position AI as a critical evaluator. They can encourage students to engage with AI-generated outputs by critiquing them and make necessary revisions based on their personal interests and needs to contextualize the outcomes (Kim et al., 2022). Moreover, when teachers review and refine AI-generated feedback, its effectiveness can be enhanced while simultaneously modelling reflective and evaluative thinking processes for students (Lo et al., 2025). Guided prompting strategies can also be incorporated into instruction, helping students learn how to request critique from AI rather than simply asking for answers, and how to contextualize prompts to obtain higher-quality outputs that reflect their own contexts while using AI (Qian, 2025). Through these approaches, students can ask AI to evaluate the clarity or logic of their draft and generate more relevant responses aligned with the topic, and their assignment goals.

Reflective activities can be integrated into instruction, guiding students to reflect on AI's contributions, its limitations, and how they accept, reject, or modify AI-generated suggestions (Tekir, 2026). It is important to maintain students' cognitive effort and productive struggle in the learning process by preventing AI from completing entire tasks, as excessive delegation may lead to excessive cognitive offloading and reduced metacognitive engagement (Bauer et al., 2025; Fan et al., 2025). Certain steps or tasks in the learning process should therefore be completed by students without AI support. For example, teachers can guide students to draft the first version of an essay or resume independently and then use AI to receive feedback afterward. These design approaches may help strengthen students' AI literacy and metacognitive skills, supporting their reflective and responsible AI use.

Consistent with the instructional approaches, assessment should address both learning processes and outcomes in AI-integrated learning contexts. Rather than focusing solely on students' final products, such as resumes or essays, teachers should evaluate how students engage in the learning process, including how they interact with AI, revise AI-generated outputs, and make decisions during the writing process (Zhan et al., 2022). To support this process- and product-oriented assessment, teachers can guide students to document their prompts, AI responses, and revisions, along with oral or written justification tasks. Teachers can also design assessments for students to explain why they revised particular wording or structures and to demonstrate their understanding of the key concepts. Students may explain the core structure of a resume or how they would adapt it to different job descriptions. Such assessment approaches can help maintain students' agency and mitigate the risk of overreliance on AI.

6. Limitation

Because this research was conducted as a single-case study involving one classroom context and a relatively small sample, the quantitative survey findings should be interpreted as descriptive indicators of students' perceptions rather than as evidence intended for statistical generalization. Instead, their primary purpose is to provide additional context that complements the qualitative analysis of student–AI interactions.

The post-survey data were collected anonymously to encourage candid student responses, which meant that individual survey responses could not be linked to specific interaction logs or reflections. Such cross-referencing would enable a more nuanced examination of how students' perceptions of AI relate to their interaction behaviors, levels of engagement, and learning processes, providing deeper insight into why some students engage critically with AI feedback while others adopt more passive approaches. Future research could build on these findings by linking students' interaction data with their survey responses.

Another limitation relates to the geographic and demographic context of the participating school. The study was conducted in a Midwestern high school classified as a town locale, referring to schools located in small urban clusters outside major metropolitan areas. Because the school represents a small-town context, the findings may reflect regional socioeconomic conditions, digital infrastructure, and students' exposure to emerging technologies such as GenAI. These contextual factors may influence students' familiarity with AI tools, access to reliable internet connectivity, and availability of technological resources. Therefore, the findings should be interpreted cautiously when considering broader generalization to urban, suburban, or highly rural settings. Future research should examine AI-supported learning across diverse geographic contexts to better understand how differences in infrastructure, socioeconomic conditions, and educational resources shape students' interactions with AI and their perceptions of its usefulness in learning.

7. Conclusion

The findings of this study revealed the key aspects of AI-integrated learning: the educational value of GenAI hinges not on the technology itself, but on students' capacity to critically and intentionally engage with it. The study suggests that AI can expand learners' agency, reflective thinking, and professional readiness only when instructional designs deliberately cultivate AI literacy, ethical awareness, and thoughtful human–AI collaboration. These insights point to an urgent need for future research that investigates how to design learning environments that foster sustained, autonomous engagement with AI; how students develop AI literacy over time across different subjects; and how schools can establish developmentally appropriate frameworks that address authenticity, ownership, and responsible use. As AI continues to reshape K–12 education, further inquiry must examine how to ensure that these tools enhance—not replace—students' cognitive growth, creativity, and sense of personal accountability.

CRediT authorship contribution statement

Kyungbin Kwon: Writing – review & editing, Writing – original draft, Supervision, Data curation, Conceptualization. **Sohheon Yang:** Writing – original draft, Methodology, Data curation. **Ugur Kale:** Writing – review & editing, Writing – original draft, Supervision. **Keunjae Kim:** Writing – review & editing, Formal analysis, Data curation. **Juhyun Park:** Writing – review & editing, Formal analysis, Data curation.

Declaration of competing interest

The authors declare the following financial interests/personal relationships which may be considered as potential competing interests:

Kyungbin Kwon reports financial support was provided by US Department of Defense. If there are other authors, they declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

This material is based upon work supported by the Department of Defense National Defense Education Program under Grant No. W15QKN-20-1-1000.

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.caeai.2026.100588.

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